

Machine Learning Capstone

Building and Comparing Course Recommender Systems

Final Assignment, Machine Learning Capstone course

IBM Machine Learning Professional Certificate

Introduction

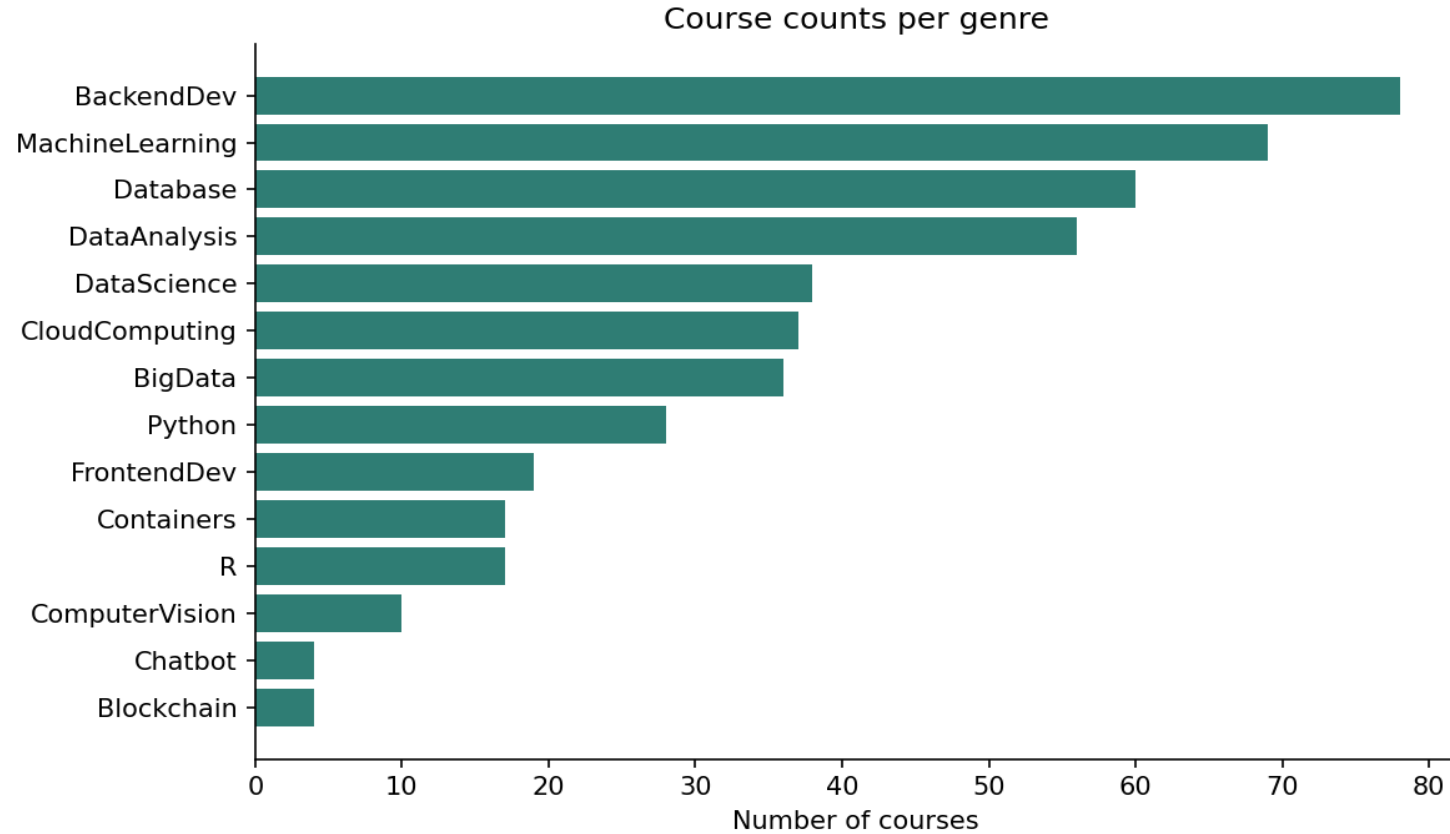
This project builds and compares recommender systems for an online course platform. The catalog has 307 courses. 33,901 learners generated 233,306 course ratings.

The goal is to recommend courses a learner has not taken yet. Six approaches are compared, in two families:

- **Content-based and clustering models:** content-based on user profile and genres, content-based on course similarity, and clustering-based. They use course genres, descriptions, or profile vectors, not other users' ratings.
- **Collaborative filtering models:** KNN, NMF, and a neural network with learned embeddings. They use only the user-item rating matrix.

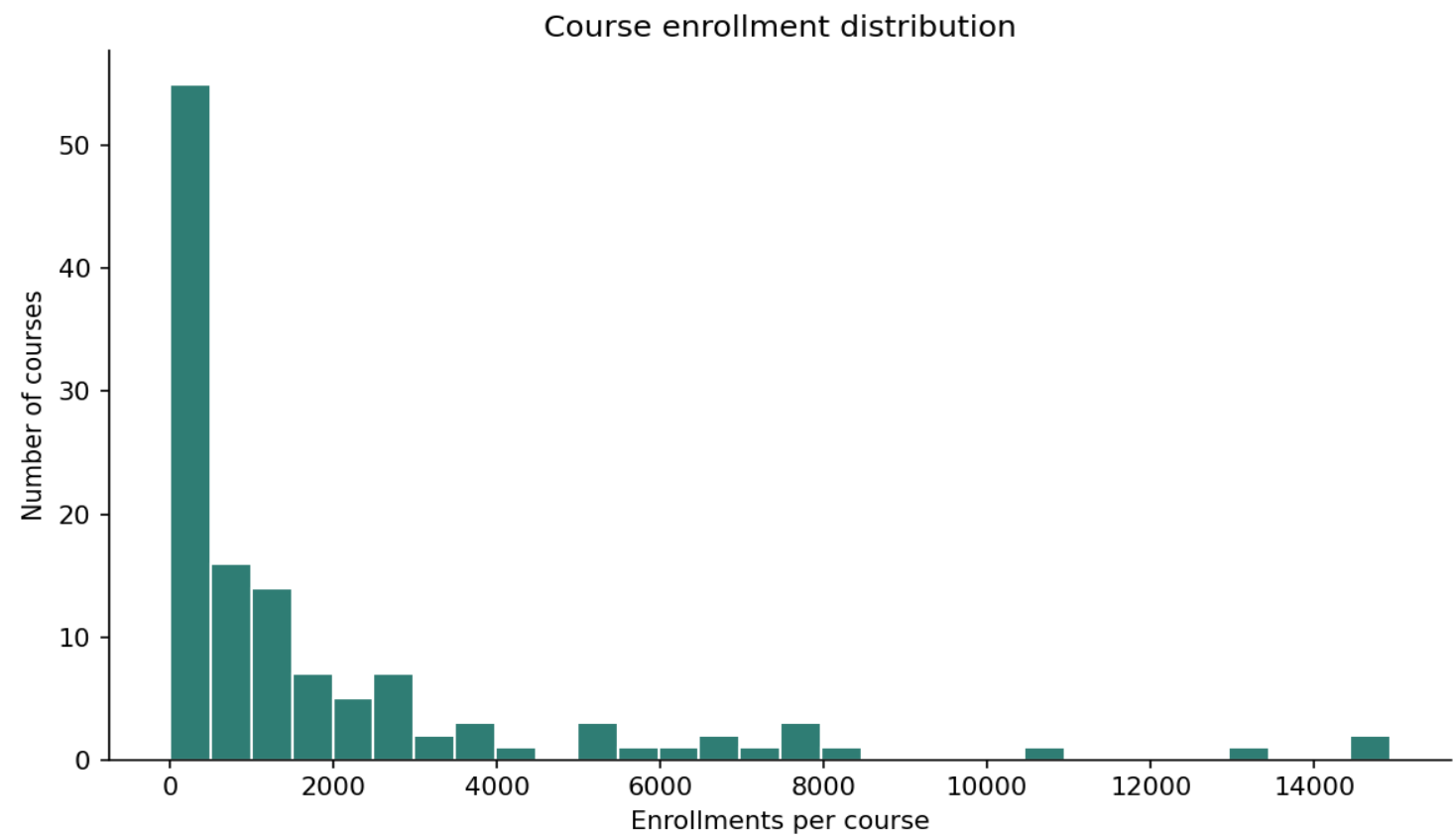
The first family is judged by how many courses it recommends, and which ones. The second family is judged by how accurately it predicts a rating, using RMSE.

Course Counts per Genre



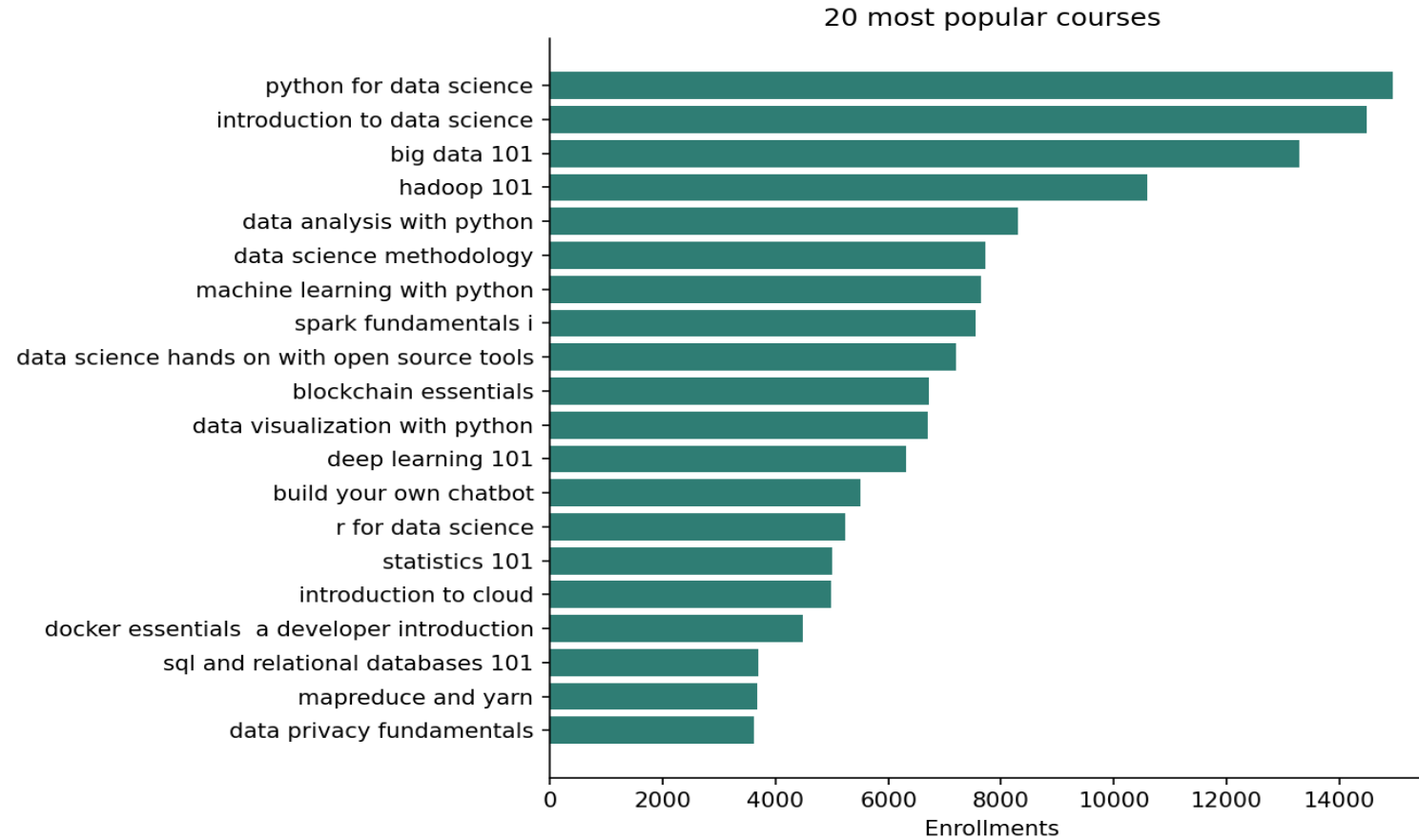
A course can belong to several genres at once, so counts do not sum to the total course count.

Course Enrollment Distribution

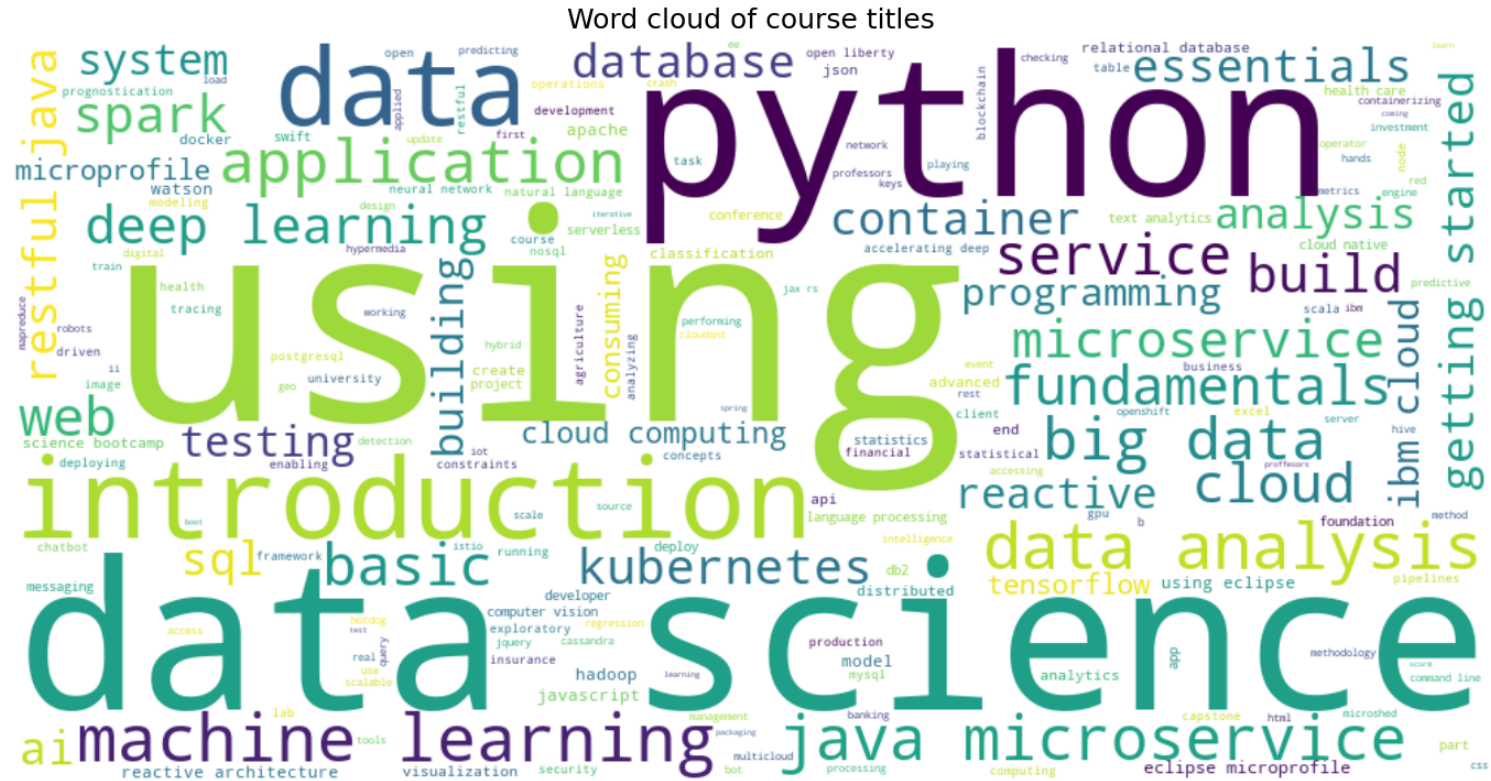


126 of 307 courses have recorded enrollments. Mean: 1852, median: 634, range: 1-14936.

20 Most Popular Courses

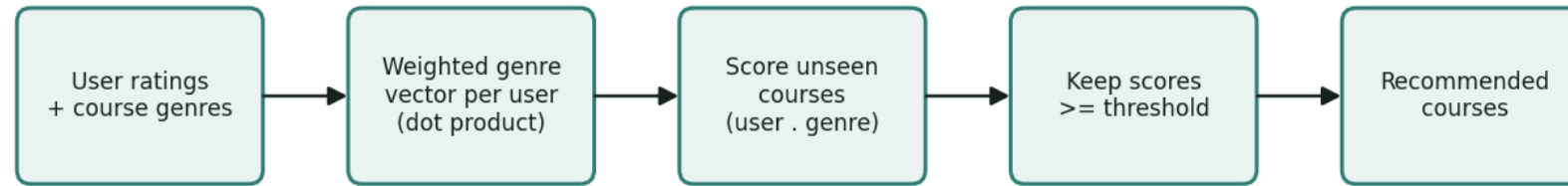


Word Cloud of Course Titles



Content-Based Recommender: User Profile and Course Genres

How it works



A user profile vector comes from the dot product of a user's rated courses and the course genre matrix. This gives a weighted interest score per genre. Unseen courses are scored the same way. Courses scoring above a threshold get recommended.

Evaluation Results: User Profile-Based Recommender

Metric	Value
Test users	1000
Score threshold	20.0
Total recommendations generated	16,710
Average recommended courses per user	28.13

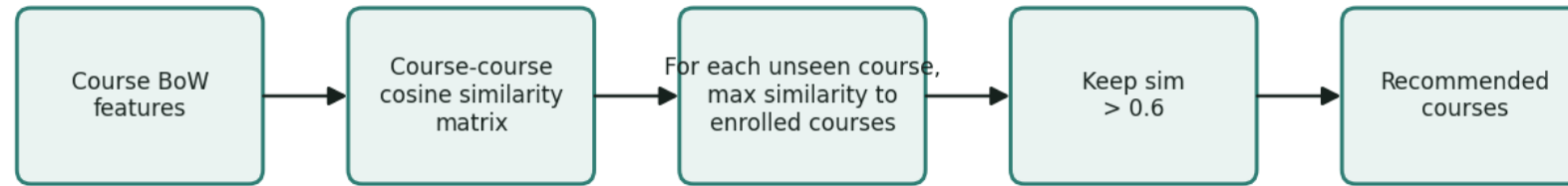
Top 10 most frequently recommended courses:

- excourse72: recommended to 322 users
- excourse73: recommended to 322 users
- TMP0105EN: recommended to 318 users
- RP0105EN: recommended to 310 users
- SC0103EN: recommended to 292 users
- excourse31: recommended to 283 users
- excourse21: recommended to 271 users
- excourse22: recommended to 271 users
- BD0212EN: recommended to 268 users
- ML0122EN: recommended to 268 users

The default threshold of 10.0 produced 61.8 recommendations per user on average, well above a usable list size. The threshold was tuned up to 20.0, bringing it down to a more reasonable range.

Content-Based Recommender: Course Similarity

How it works



Each course was already converted to a bag-of-words vector from its description. A course-to-course cosine similarity matrix was precomputed from these vectors. An unseen course is recommended if its similarity to any enrolled course exceeds 0.6.

Evaluation Results: Course Similarity Recommender

Metric	Value
Similarity threshold	0.6
Average recommended courses per user	11.38

Top 10 most frequently recommended courses:

- excourse22: recommended to 579 users
- excourse62: recommended to 579 users
- DS0110EN: recommended to 562 users
- excourse63: recommended to 555 users
- excourse65: recommended to 555 users
- excourse72: recommended to 551 users
- excourse68: recommended to 550 users
- excourse74: recommended to 539 users
- excourse67: recommended to 539 users
- BD0145EN: recommended to 506 users

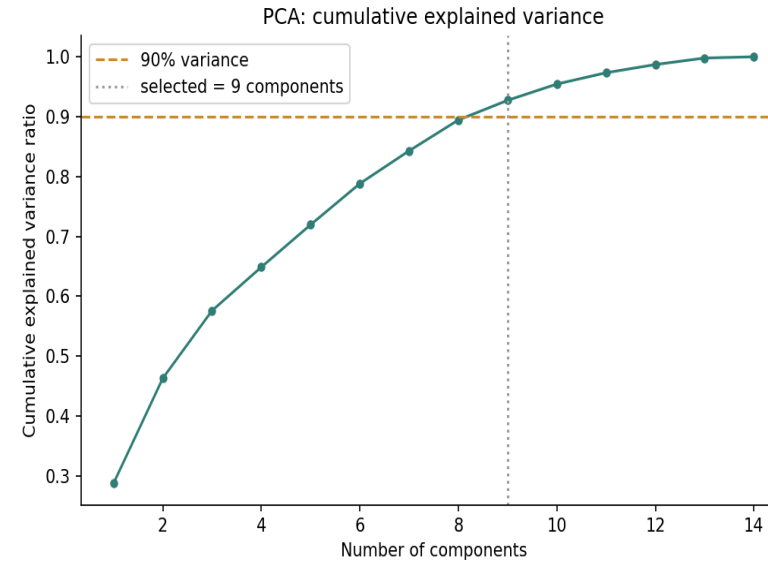
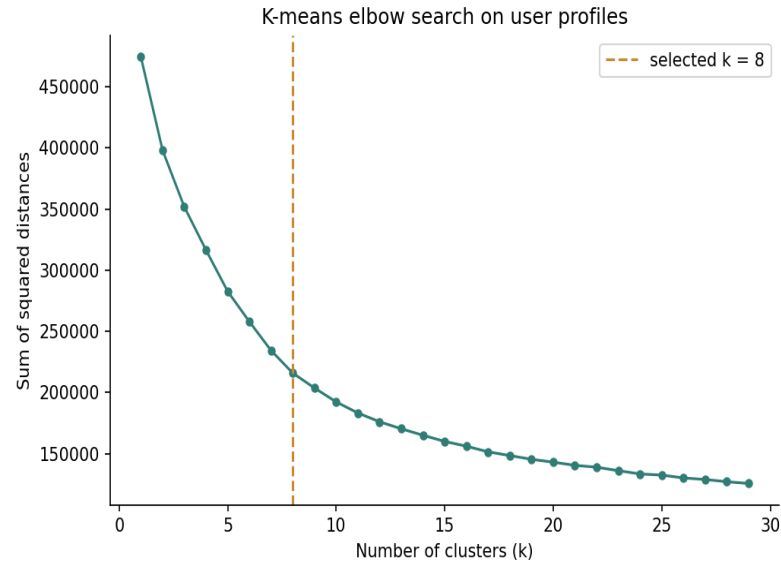
Clustering-Based Recommender

How it works



Users are grouped by learning interest with K-means on their standardized profile vectors. Each user gets the popular courses in their own cluster, if they have not taken them yet. A course counts as popular if at least 10 cluster members took it.

Clustering: Choosing k and PCA Components



Elbow method selects $k = 8$ clusters. PCA needs 9 components to keep 92.7% of the variance in the 14 original genre-interest features.

Evaluation Results: Clustering-Based Recommender

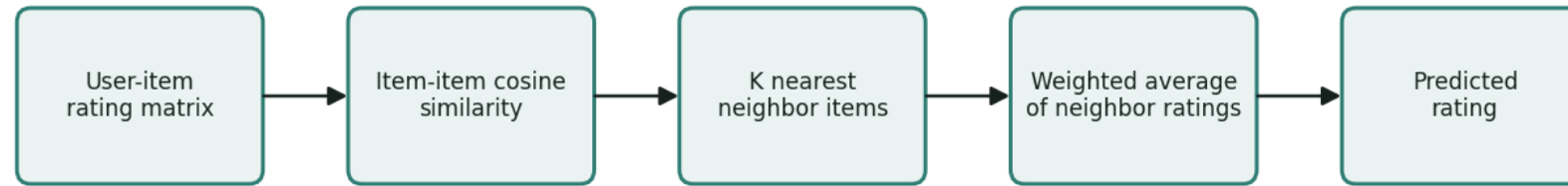
Metric	Value
Number of clusters (k)	8
PCA components used	9
Cluster enrollment threshold	10
Average recommended courses per user	28.63

Top 10 most frequently recommended courses:

- DB0101EN: recommended to 834 users
- DS0301EN: recommended to 829 users
- CO0101EN: recommended to 783 users
- ST0101EN: recommended to 776 users
- RP0101EN: recommended to 750 users
- BC0101EN: recommended to 723 users
- CL0101EN: recommended to 716 users
- ML0115EN: recommended to 708 users
- ML0101EN: recommended to 708 users
- WA0101EN: recommended to 703 users

Collaborative Filtering: KNN-Based Recommender

How it works

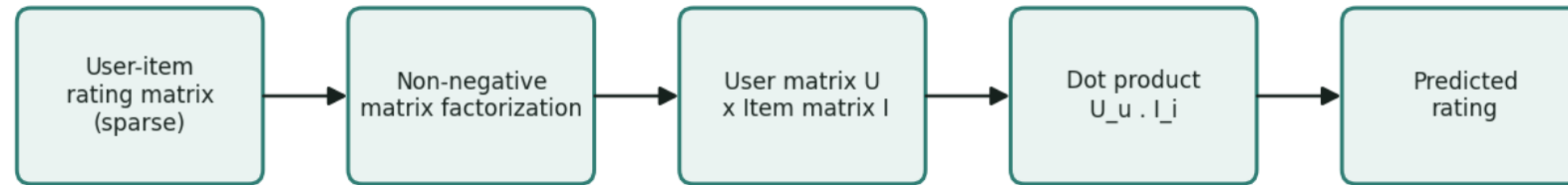


Item-based KNN computes cosine similarity between courses directly from the user-item rating matrix (no genres or descriptions needed). A rating is predicted as the similarity-weighted average of a user's ratings on the course's nearest neighbors.

Implementation: surprise KNNBasic, item-based cosine similarity, min_k=5, max_k=100, 70/30 train-test split. Test RMSE: 0.2081.

Collaborative Filtering: NMF-Based Recommender

How it works

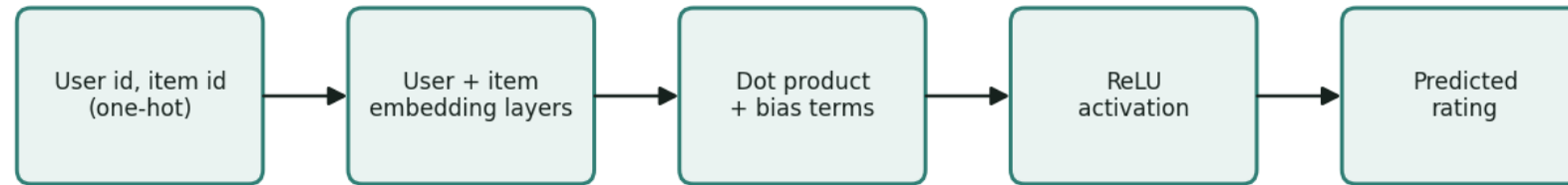


Non-negative matrix factorization decomposes the sparse user-item matrix into a dense user factor matrix and a dense item factor matrix. Multiplying a user's row by an item's column predicts the rating. Unlike KNN, no full similarity matrix needs to be kept in memory.

Implementation: surprise NMF, same 70/30 train-test split. Test RMSE: 0.2097.

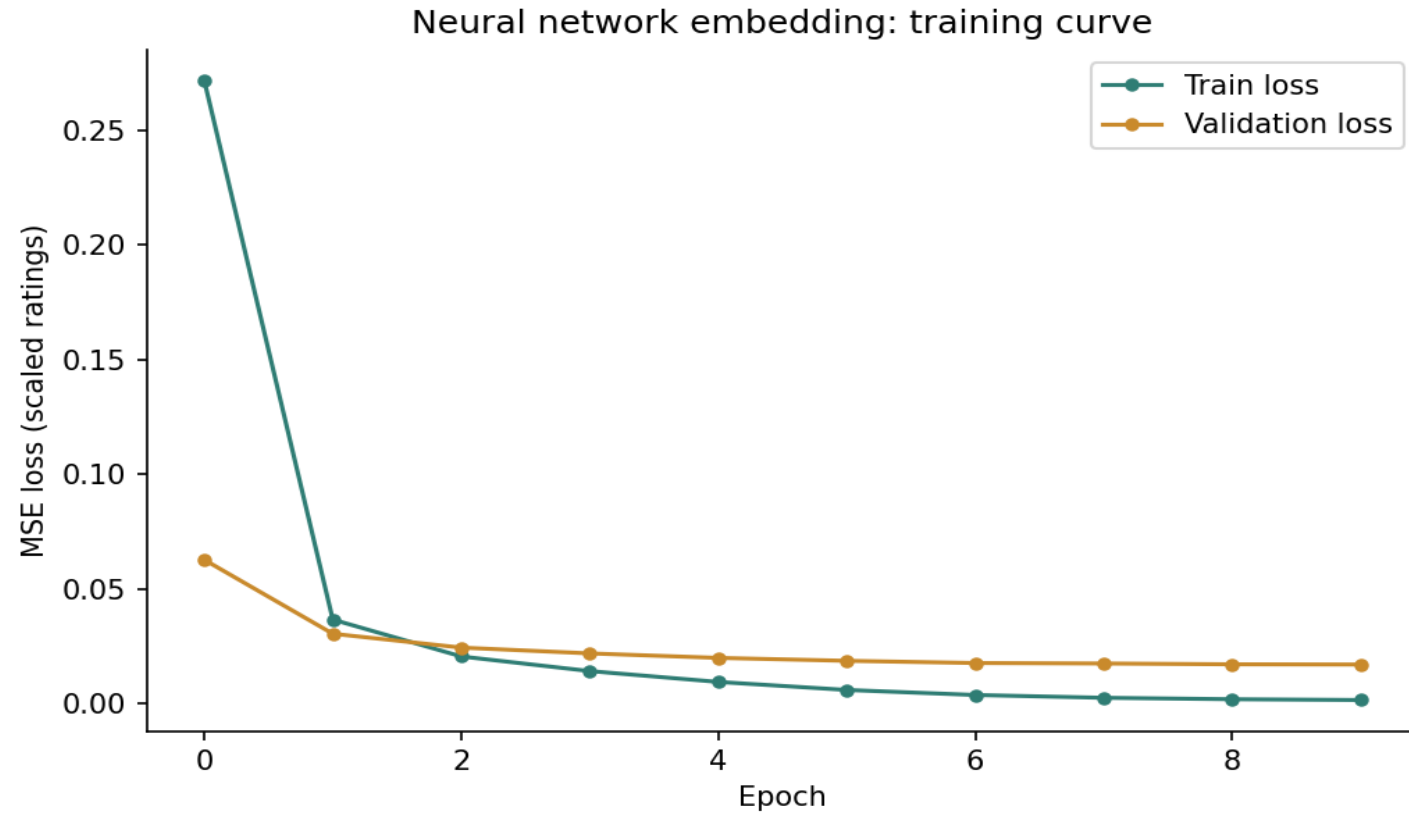
Collaborative Filtering: Neural Network Embedding-Based Recommender

How it works



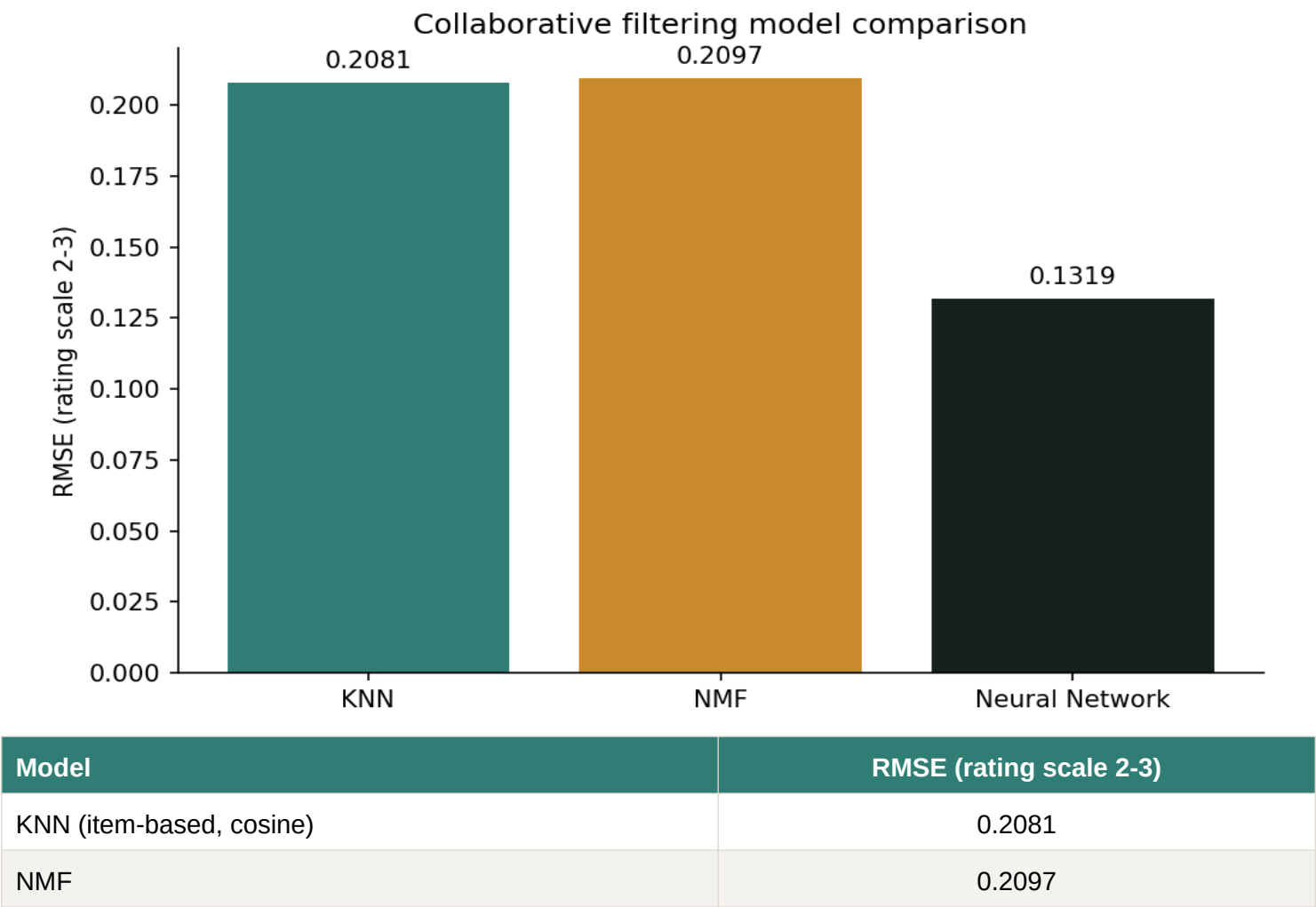
A neural network learns a latent embedding vector and a bias term for each user and each course. The predicted rating is the dot product of the user and item embeddings, plus both biases. A ReLU activation is applied last.

Neural Network: Training Curve



Embedding size: 16. Trained for 10 epochs, batch size 64, Adam optimizer, MSE loss on ratings scaled to [0, 1]. Test RMSE (original rating scale): 0.1319.

Model Performance Comparison



Neural network (embeddings)	0.1319
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Conclusion

The neural network had the lowest RMSE of the three collaborative filtering models (0.1319). It was the most accurate at predicting exact ratings in this comparison.

- Content-based and clustering models are easy to explain. A recommendation always traces back to a shared genre, a similar course description, or a shared interest cluster.
- Collaborative filtering models need no course content, only rating history. Their recommendations are harder to explain to a learner.
- A production system could combine both. Collaborative filtering would be the main engine for users with enough history. Content-based would cover new users and new courses, where no rating history exists yet.